

Seyed Abolfazl Mortazavi | Data Scientist | AI Engineer

Tehran, Iran

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Professional Summary

Data Scientist and AI Engineer building production machine-learning systems across speech, NLP, computer vision, and signal processing. At Shatel, own end-to-end data and ML workflows from data collection and curation through training, evaluation, API delivery, and production integration. Experienced with large-scale speech and conversational datasets, statistical analysis, LLM-assisted text analytics, and translating technical findings into product and operational decisions.

Professional Experience

Shatel

Tehran, Iran

Data Scientist / AI Researcher - Speech Team

Jan. 2025–Present

- Own end-to-end data and ML workflows for production speech, text, and document-intelligence systems; processed and curated **hundreds of thousands of hours of speech data** and millions of conversational transcripts, with approximately **50,000 hours** used for model training.
- Developed and evaluated ASR systems achieving **2.97% WER** on evaluated Persian speech, **8.02% WER** on noisy call-center audio, and bilingual results of **4.02% Persian WER** and **2.06% English WER**.
- Independently designed and delivered a production call-center quality-control and analytics system from data preparation through API integration, processing **1,000+ calls/day** across speech recognition, acoustic emotion analysis, transcript analytics, conversation-quality indicators, and automated summarization.
- Analyzed call-center operations and identified recording-quality, call-drop, and process issues; presented evidence to stakeholders and informed corrective-action planning.
- Built a six-class speech-emotion component for noisy call-center audio, achieving **87% Macro-F1**, and developed an end-of-utterance workflow from dataset preparation through evaluation and integration.
- Built preprocessing, text-detection/localization, evaluation, and FastAPI layers for multilingual Persian/Arabic/English OCR, achieving **>97% F1** on text localization.
- Developed and evaluated a **16-bit speech-watermarking** pipeline for synthesized speech, achieving **0 BER on clean audio** and **<1% average BER** across evaluated perturbations including cropping, noise, resampling, compression, speed, and pitch changes.
- Prepared technical reports and presented AI-system results and product capabilities in technical, product, and presales discussions.

Independent Contractor

Freelance AI & Signal Processing Engineer

Nov. 2023–Present

- Deliver confidential machine-learning, signal-processing, and computer-vision projects under NDA, covering data preparation, analysis, predictive modeling, and deployment in Python and MATLAB.

Technical Skills

Data Science: Python, MATLAB, NumPy, Pandas, SciPy, Scikit-learn, XGBoost; EDA, feature engineering, model evaluation

ML / Deep Learning: Classification, regression, PyTorch, PyTorch Lightning, TensorFlow, Hugging Face Transformers

Statistics: Hypothesis testing (e.g., Chi-square and Kolmogorov–Smirnov), statistical model evaluation

Speech / Audio: ASR, VAD, EOT, speech emotion recognition, speaker identification / verification, audio preprocessing, enhancement, augmentation, Librosa

NLP / LLMs: Text classification, sentiment analysis, conversation analysis, summarization, prompt engineering, transformer-based models

Computer Vision: Image classification, segmentation, object detection, OCR / text localization, image registration, preprocessing / enhancement, OpenCV

MLOps / Backend: MLflow, TensorBoard, FastAPI, Docker, Git/GitHub, Linux, Conda

Research Experience

Computational Neuroscience Lab, Amirkabir University of Technology

Tehran, Iran

Research Assistant & Thesis Mentor

Apr. 2025–Present

- Mentor six students (1 B.Sc., 4 M.Sc., 1 Ph.D.) on machine learning, neural signal processing, data analysis, statistics, implementation, and research methodology; two completed B.Sc. theses received **20/20** and **19/20**.

Computational Neuroscience Lab, Amirkabir University of Technology

Tehran, Iran

BCI Researcher

Aug. 2022–Oct. 2023

- Developed quantitative EEG signal-quality measures and evaluated research hypotheses using MATLAB/Python analysis and statistical tests including Chi-square and Kolmogorov–Smirnov.

Education

Sharif University of Technology

Tehran, Iran

M.Sc. Electrical Engineering - Bioelectric, GPA: **19.02/20**; Rank: **1st/17**

2023–2025

Thesis: Eye Tracking Using EEG Signals During Simultaneous Functional Magnetic Resonance Imaging.

Supervisor: Dr. Ali Ghazizadeh Ehsaei **Thesis grade:** **20/20 (Excellent)**

Amirkabir University of Technology

Tehran, Iran

B.Sc. Biomedical Engineering - Bioelectric, GPA: **19.07/20**; Rank: **1st/121**

2019–2023

Additional studies: Concurrent double-major coursework in Electrical Engineering (Telecommunications), Amirkabir University of Technology; GPA **19.38/20**, Rank **1st/132**. The second-degree track was discontinued after admission to the M.Sc. program, with approximately 20 credits remaining.

Selected Projects

Image Classification: Built a PyTorch computer-vision pipeline achieving **98.37% accuracy**. [GitHub](#)

Speech Enhancement: Implemented DSP-based speech-enhancement and noise-reduction algorithms in MATLAB. [GitHub](#)

NLP: Developed machine-learning approaches for fake-news classification. [GitHub](#)

Teaching Experience

Fall 2025: Teaching Assistant, Blind Source Separation - Amirkabir University of Technology

Fall 2024: Teaching Assistant, Machine Learning - Sharif University of Technology

Fall 2024: Teaching Assistant, Neural Modeling - Sharif University of Technology

Fall 2022: Teaching Assistant, Engineering Mathematics - Amirkabir University of Technology

Professional Development

Neuromatch Academy Deep Learning; Deep Learning & Large Language Models (Sharif OCW); Advanced Learning Algorithms (Coursera); Complete Neural Signal Processing and Analysis; EEG Signal Processing

Languages

Persian: Native

Turkish: Native

English: Professional working proficiency